The CTI Universal Law

A First-Principles Visual Guide

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The Big Idea

There is a **single equation** governing
how well *any* representation separates categories.

$$\mathrm{logit}(q) = \alpha\,\kappa - \beta \log(K-1) + C$$

Derived from first principles.
Validated across NLP, vision, audio, and biological cortex.

What Are We Measuring?

The question: How good are a model's internal representations at telling categories apart?

The probe: 1-Nearest Neighbor (1-NN)

- Find closest training point
- Predict same label
- No learnable parameters
- Accuracy = pure geometry

Why not a linear probe?

Linear probes mix geometry with optimization. 1-NN removes this confounder: **accuracy IS geometry**.

Key Quantity #1: q_{norm}

Chance-Normalized Accuracy

$$q_{\text{norm}} = \frac{\text{acc}_{\text{1NN}} - 1/K}{1 - 1/K}$$

- $q = 0$: at chance (no signal)
- $q = 1$: perfect classification
- Comparable across different K

Example: K=10, acc = 55% $\rightarrow q = (0.55 - 0.10)/(0.90) = 0.50$

Key Quantity #2: κ_{nearest}

The Geometric Signal-to-Noise Ratio

$$\kappa_{\text{nearest}}(k) = \min_{j \neq k} \frac{\|\boldsymbol{\mu}_j - \boldsymbol{\mu}_k\|}{\sigma_W \sqrt{d}}$$

Numerator:
Distance to the **nearest** other class centroid (hardest competitor)

Denominator:
Expected within-class L2 radius (noise level)

Why "nearest"?

In a 1-NN race, you confuse with the **nearest** class. Far classes are irrelevant. This is the **bottleneck**.

κ Intuition: Signal vs. Noise

κ	Regime	Meaning
$\ll 1$	Noise dominates	Classification at chance
~ 1	Signal $\approx$ noise	Transition zone
$\gg 1$	Signal dominates	Near-perfect

κ is **dimensionless** — comparable across models with different embedding dimensions.

Think of it like a "contrast ratio" for class separation.

The Gumbel Race: Step 1 — 1-NN as a Race

The Competition

$$M_+ = \text{distance to nearest } \textbf{same-class} \text{ point}$$
$$M_j = \text{distance to nearest point from class } j$$

$$\text{1-NN is correct} \iff M_+ < \min_{j \neq k} M_j$$

The correct class must **win the race** by having the smallest minimum distance.

The Gumbel Race: Step 2 — Extreme Value Theory

Fisher–Tippett theorem (1928): The minimum of n i.i.d. random variables converges to one of three universal distributions:

1. **Gumbel** — exponential tails (Gaussian, etc.) $\leftarrow$ **Our case**
2. Fréchet — heavy tails
3. Weibull — finite support

After EVT normalization

$$M_+ \sim \mathrm{Gumbel}(\mu_+, s)$$

$$M_j \sim \mathrm{Gumbel}(\mu_j^{(-)}, s) \quad \text{shared scale } s$$

The Gumbel Race: Step 3 — The Magic Identity

The Gumbel Race Identity (EXACT)

If $G_0, G_1, \ldots, G_{K-1}$ are independent Gumbels with common scale s:

$$\boxed{P(G_0 < \min_{j \geq 1} G_j) = \frac{1}{1 + \sum_j \exp\big(-(\mu_j - \mu_0)/s\big)}}$$

This is **exact**. Not an approximation.

The probability is a **logistic function** of the location gaps.

This is why the law has a sigmoid/logit form.

The Gumbel Race: Steps 4–5 — Competition Structure

Apply to 1-NN:

$$q = \frac{1}{1 + \sum_{j \neq k} \exp(-\Delta_{k,j}/s)}$$

Two regimes:

Dense (ETF)

All $K-1$ competitors equal:
$\sum = (K-1)e^{-\Delta/s}$
$\Rightarrow \beta = 1$

Hierarchical (Real World)

Only $\sqrt{K}$ active competitors:
$\sum = \sqrt{K-1}\, e^{-\Delta/s}$
$\Rightarrow \beta = 0.5$

The Gumbel Race: Steps 6–7 — The Final Law

Gap is linear in κ: $\Delta_{\text{nearest}}/s \approx a_1 \kappa_{\text{nearest}} + a_0$

Combine:

The CTI Universal Law

$$\text{logit}(q_{\text{norm}}) = \underbrace{\alpha}_{\text{slope}} \kappa_{\text{nearest}} - \underbrace{\beta}_{\text{competition}} \log(K-1) + \underbrace{C}_{\text{offset}}$$

Derived (theorem):

- Logistic in κ
- Logarithmic in K

Estimated (data):

- $\alpha = 1.477$ (NLP)
- $\beta = 0.478$
- C per dataset

Neural Collapse: Why It Works Across Architectures

Neural Collapse (Papyan et al., PNAS 2020)

At end of training, representations converge to:

1. **NC1**: Within-class variability collapses
2. **NC2**: Centroids form equiangular tight frame (regular simplex)
3. **NC3**: Classifier aligns with centroids
4. **NC4**: Predictions $\rightarrow$ nearest-centroid

κ_{nearest} measures **proximity to Neural Collapse**.

Different architectures $\rightarrow$ similar NC geometry $\rightarrow$ same law.

Architecture determines the **path**; law governs the **endpoint**.

The Equicorrelation ρ: Tightest Invariant

ρ = avg whitened cosine similarity between centroid-difference vectors.

Modality	$\boldsymbol{\rho}$
NLP decoders	0.463
Audio speech	0.455–0.464
Vision (ViT, CNN)	0.458–0.467
Mouse V1 cortex	0.466
NLP encoders	0.446–0.456
All 6 modalities	**0.462**

Key Facts

- CV = 1.0% (tightest invariant!)
- Simplex prediction: $\rho = 0.5$
- Observed: 8% below perfect NC
- **Substrate-independent**

α prediction

$$\alpha_{\text{pred}} = \sqrt{4/\pi}/\sqrt{1-\rho} = 1.534$$

Observed: 1.477 (+3.8%)

α by Signal Type

Signal Type	α	Modalities
Discrete tokens	$\sim$1.48	NLP decoders (CV = 2.3%)
Continuous signals	$\sim$3.3–5.0	ViT (4.5), CNN (4.0), Audio (4.7)
Encoders	$\sim$7.7	BERT, DeBERTa, ELECTRA

- Within NLP decoders: α is a **true constant** (CV consistent with estimation noise)
- Across families: varies by signal type (NOT a failure — level-2 universality)
- All continuous-signal modalities converge at $\alpha \approx$ 4–5
- **Open problem**: why the gap? Same ρ, different $\alpha \to \Sigma_W$ structure

Cross-Modal Validation

Modality	K	α	Result	Status
NLP (12 arch.)	4–28	1.477	$R^2 = 0.955$	**PASS**
ViT-Large	10	4.5	$R^2 = 0.964$	**PASS**
ResNet50 CNN	100	4.4	$r = 0.749$	**PASS**
Audio (7 models)	36	4.669	$r = 0.898$	**PASS**
Mouse V1 (32 mice)	118	—	30/32	**PASS**
Protein LMs	7	-1.17	$r = -0.15$	**FAIL**
Human fMRI	12	—	$r = 0.12$	**FAIL**

Form universality: NLP, vision, audio, biological cortex. Same equation.

Honest scope: protein LMs and fMRI fail (documented in paper).

The Generation Law

Key Insight

Next-token prediction $=$ V-way classification. Same Gumbel race. κ from unembedding matrix W_U (zero forward passes!) predicts perplexity.

Test	$r(\kappa, \log \mathrm{CE})$	**Note**
All 22 models	-0.55 ($p = 0.008$)	4 arch types
Excl. LFM outlier (21)	-0.70 ($p < 0.001$)	
Fixed-V Pythia+Mamba (10)	-0.837 ($p = 0.003$)	Arch-independent

Vocabulary drops out: $\beta_{\ldots} \approx 0$ because $K_{\ldots} \approx 2, 3$ (not V)

Causal Evidence: 5 Tiers

1. **Frozen do-interventions**: Move centroid $\rightarrow$ nearest $r = 0.899$; farthest $r = 0.000$
2. **Orthogonal factorial**: Arm A $r = 0.899$; Control $r = 0.000$
3. **Cross-modal dose-response**: 6 unseen architectures, 6/6 PASS
4. **Confusion matrix prediction (pre-registered, zero refit):**

δ	r	Sign accuracy	p-value
1.0	0.842	92.9%	10^{-50}
2.0	0.808	**100%**	10^{-43}
3.0	0.776	**100%**	10^{-38}

5. **Top-m sweep**: m=2 recovers LOAO α within 2.3%

Three-Level Universality

Level 1: Form (strongest)

Logit-linear structure is EVT-derived. Confirmed across NLP, vision, audio, biology.

Analogy: $PV = nRT$ holds for all ideal gases.

Level 2: Constant within signal type

α stable within families (NLP decoders: CV $= 2.3\%$). Varies across signal types.

Analogy: van der Waals a, b differ by substance.

Level 3: Intercept is task-specific

C encodes dataset difficulty. One calibration measurement required per

Open Directions

Near-term:

- Expanded generation law (25+ models)
- External replication
- COLM 2026 submission (Mar 26/31)

Open theory:

- **Renormalization**: Why $\alpha_{\mathrm{NLP}} \neq \alpha_{\mathrm{ViT}}$ with same ρ?
- Non-asymptotic bounds (Berry–Esseen for Gumbel)
- Multilabel extension

Long-term:

- Training objective: maximize $\kappa_{\mathrm{nearest}}$
- Connection to scaling laws: $q(C) = \sigma(\alpha C^{\gamma} + C_0)$
- Universal constant from first principles

The Complete Derivation in 7 Steps

1. **Define** $\kappa_{\text{nearest}}(k) = \min_{j \neq k} \|\boldsymbol{\mu}_j - \boldsymbol{\mu}_k\| / (\sigma_W \sqrt{d})$
2. **1-NN race**: correct $\iff M_+ < \min_j M_j$
3. **EVT**: $M_+, M_j \to \mathrm{Gumbel}(\mu, s)$ with shared scale
4. **Gumbel race** (exact): $P = [1 + \sum \exp(-\Delta/s)]^{-1}$
5. **Sparse hierarchy**: $\sum \approx (K-1)^{0.5} \exp(-\Delta_{\text{near}}/s)$
6. **Linearize**: $\Delta_{\text{near}}/s \approx a_1 \kappa + a_0$
7. **Combine**:

$$\text{logit}(q) = \alpha\, \kappa - \beta \log(K-1) + C$$

Derived: the functional form (logistic, logarithmic).
Estimated: the constants (α, β, C).

One equation.
Five modalities.
First principles.

$$\mathrm{logit}(q) = \alpha\, \kappa_{\mathrm{nearest}} - \beta \log(K-1) + C$$